
Exercise Sheet 3:
Constraint Satisfaction Problems

Foundations of Neuro-Symbolic AI
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May 11, 2026

n rooks problem

For arbitrary $n \in \mathbb{N}$ let there be a grid of $n \times n$ positions with n columns and n rows. We want to assign r rooks onto positions of the chess board, such that:

- Each column contains at most one rook.
- Each row contains at most one rook.

Task 1:

Design the n rooks problem as a Constraint Satisfaction Problem.

- (i) What are suitable variables, their leg dimensions, and the constraint tensors?
- (ii) Which hidden variables could be introduced for a sparser representation of the constraints?

Hint: Compare the problem to the Sudoku game.

Task 2:

Assume again that we have a partial assignment of $r \leq n$ rooks given. How many assignments of the remaining $n - r$ rooks satisfy the constraints? How to compute that number by a tensor network contraction?

Task 3 (Generalized unit propagation):

Assume that we have a partial assignment of $r < n$ rooks given. Would the generalized unit propagation algorithm be able to find a solution for the remaining $r - k$ rooks?

Task 4 (Backtracking):

Assume that we have a partial assignment of $r < n$ rooks given. How would the backtracking algorithm with generalized unit propagation as a simplifier find a solution for the remaining $r - k$ rooks?